

Møller Polarimetry for PV Experiments at 12 GeV

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MOLLER Review

Outline

- 1 Introduction
- 2 Ongoing Upgrade in Hall A
- 3 Møller with Atomic Hydrogen Target
- 4 Conclusion

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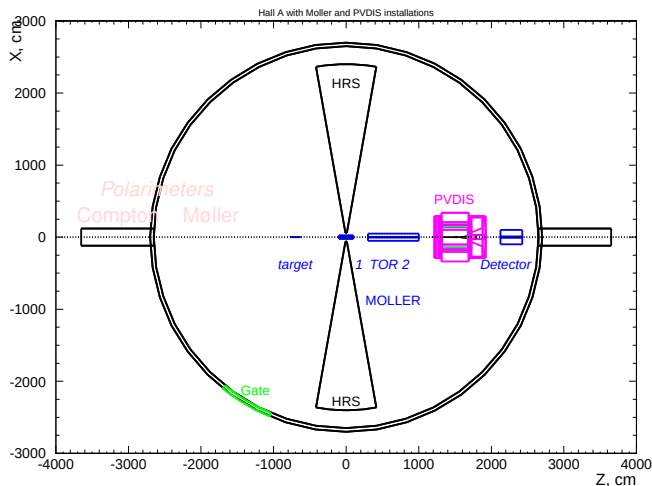
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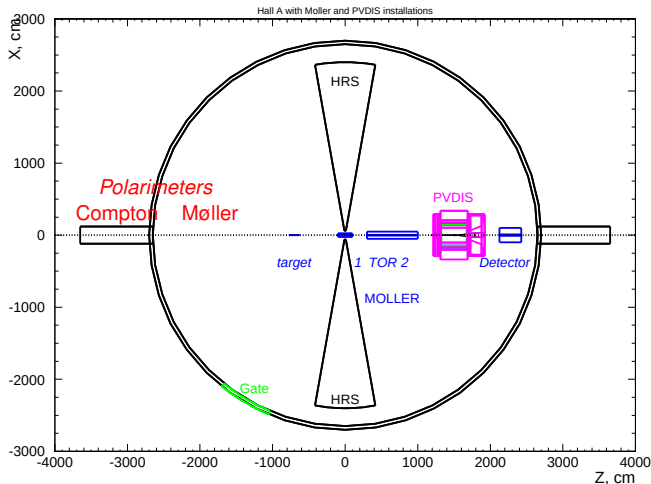
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Installation in Hall A



Installation in Hall A



Error Budget of Møller and SoLID Experiments

Møller

Source of error	% error
Q^2 absolute value	0.5
beam polarization	0.4
beam second order	0.4
inelastic ep	0.4
elastic ep	0.3
other	0.5
total	1.0

SoLID

Source of error	% error
beam polarization	0.4
radiative corrections	0.3
Q^2 absolute value	0.2
statistics	0.3
total	0.6

0.4% - can it be done?

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Electron Polarimetry for PV at JLab: Features

- Energy range $E_{beam} = 6.6 - 11 \text{ GeV}$
- Current range $I_{beam} = 40 - 90 \mu\text{A}$

Additional features to consider

- Time needed to achieve $\sim 0.4\%$ statistical error
- Systematic error:
 - Does polarimetry use the same beam as the experiment (energy, current, location)?
 - Continuous or intermittent (invasive?)
- Two different polarimeters/methods highly desirable

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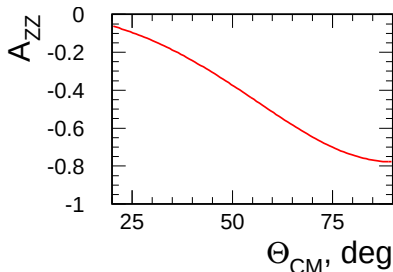
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Møller Polarimetry

- Rad. corrections to Born $< 0.3\%$
- Detecting the e^- at $\theta_{CM} \sim 90^\circ$
- $\frac{dA}{d\theta_{CM}}|_{90^\circ} \sim 0$ - good systematics
- Beam energy independent
- Coincidence - no background
- Ferromagnetic target $\mathcal{P}_T \sim 8\%$
 - Heating $\frac{dP}{dT} \sim 1\%/100^\circ\text{C}$
 $\langle I_B \rangle < 3 \mu\text{A}$
 - Levchuk effect (atomic e^-)
 - Low $\mathcal{P}_T \Rightarrow$ dead time
 - Syst. error $\sigma(\mathcal{P}_T) \sim 2\%$ (0.5%?)

$$\vec{e}^- + \vec{e}^- \rightarrow e^- + e^- \text{ QED.}$$



$$A(E) = -\frac{7}{9}$$

$$\sigma_{lab} \sim 180 \frac{\text{mb}}{\text{ster}}$$

Møller Polarimetry

Polarized electron targets: magnetized ferromagnetic foils

- Iron: polarized d -shell (6 positions occupied out of 10)
- $\mathcal{P}_e$ not calculable: derived from measured magnetization
- Spin-orbital corrections ($\sim 5\%$) - measured in bulk material
- Magnetizing field is along the beam

Field 20 mT, foil at $\sim 20^\circ$

- Magnetization along the foil
- Magnetization can be measured
- A few % from saturation
- Sensitive to annealing, history
- Polarization accuracy $\sim 2 - 3\%$

Field 3 T, foil at $\sim 90^\circ$

- Magnetization perp. to the foil
- Magnetization - from world data
- Foil saturated
- Polarization is robust.
- Polarization accuracy $\sim 0.5\%$

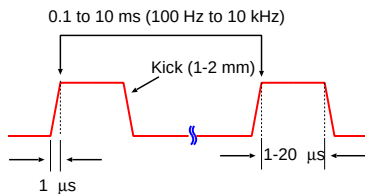
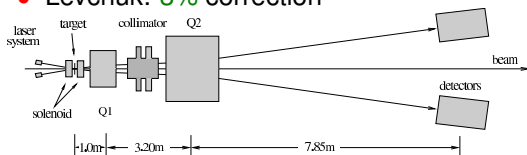
Møller Polarimeter with Saturated Iron foil (Hall C)

JLab, Hall C, M. Hauger *et al.* NIM A **462**, 382 (2001), talk on PAVI09 by S. Page

- External $B_z \sim 3 - 4 \text{ T}$
- Target foils $1\text{-}10 \mu\text{m}$, perp. to beam
- $\mathcal{P}_t$ not measured
- Levchuk: 3% correction

Tests at high beam current

- Half-moon shape foil
- Kicker magnet

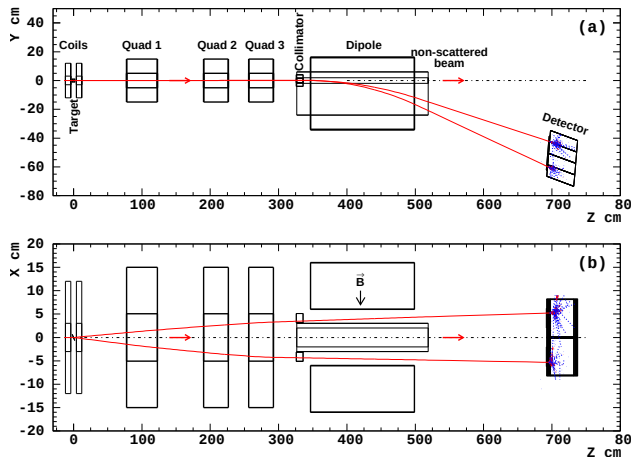


source	$\sigma(A)/A$
optics, geometry	0.20%
target	0.28%
Levchuk effect	0.30%
total at $3 \mu\text{A}$	0.46%
$\Rightarrow 100 \mu\text{A}$	?

A $1 \mu\text{m}$ thick half-foil: mech. problems:

- Foil unstable: holder design
- Thicker foil - high rate
- At $20 \mu\text{A}$ - accidentals/real ≈ 0.4

Hall A Møller Polarimeter



- Minimal Levchuk
- $\sigma_{stat} = 1\%$ in $\sim 2-3$ min
- $B_z \sim 25$ mT field
- Foil at 20° to field
- Foils $5-30 \mu\text{m}$
- Beam $< 2 \mu\text{A}$
- Systematics $\sim 2\%$

Hall A Møller Polarimeter: current upgrade

Upgrade, motivated by PREX requirements:

- High field magnetization (Hall C target clone)
- High instantaneous beam current:
reduce heating by introducing a beam duty cycle $< 5\%$
 - Beam rep. rate $500 \text{ MHz}/4 - \mathcal{F}_{laser} \cdot (n + 1) = \mathcal{F}_{RF} \cdot n$ “beat”
 - “Tune beam”: 4 ms pulses at $\sim 60 \text{ Hz}$
 - Instantaneous counting rate at $50 \mu\text{A}$ will be $\times 3$ higher
 - More invasive than a kicker scheme
- Electronics upgrade to digest higher rates

Target heating with the real raster

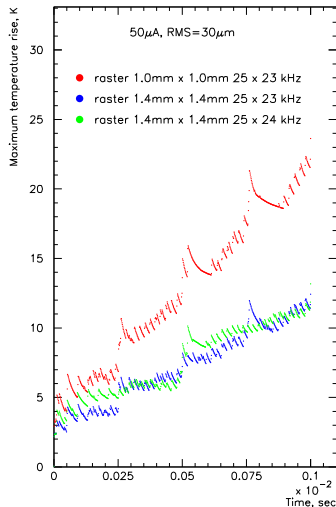
Average heating by $1.5 \mu\text{A}$

- $\Delta T_{max} \sim 22 \text{ K}$ no raster
- $\Delta T_{max} \sim 12 \text{ K}$ raster $1 \times 1 \text{ mm}^2$

Instantaneous heating by $50 \mu\text{A}$

$50 \mu\text{A}$, $\sigma_X \sim 30 \mu\text{m}$, $\Delta t = 1 \text{ ms}$
 Raster $\sim 1.4 \times 1.4 \text{ mm}^2$, $25 \times 24 \text{ kHz}$

- In pulse $\Delta T_{max} \sim 12 \text{ K}$
- Total $\Delta T_{max} \sim 24 \text{ K}$ - acceptable!



Møller Systematic Errors

Variable	Hall C	Hall A	
		Present	Upgrade
Target polarization	0.25%	2.00%	0.50%
Target angle	0.00%	0.50%	0.00%
Analyzing power	0.24%	0.30%	0.30%
Levchuk effect	0.30%	0.20%	0.20%
Target temperature	0.05%	0.00%	0.02%
Dead time	?	0.30%	0.30%
Background	?	0.30%	0.30%
Others	0.10%	0.30%	0.30%
Beam extapolation	?	larger	?
Total	0.47%	2.10%	~0.80%

Possible Breakthrough in Accuracy

Møller polarimetry with 100% polarized atomic hydrogen gas, stored in a ultra-cold magnetic trap.

E.Chudakov and V.Luppov IEEE Trans. on Nucl. Sc., 51, 1533 (2004)

http://www.jlab.org/~gen/hyd/loi_3.pdf

Advantages:

- 100% electron polarization
 - very small error on polarization
 - sufficient rates $\sim \times 0.005$ - no dead time
 - false asymmetries reduced $\sim \times 0.1$
- Hydrogen gas target
 - no Levchuk effect
 - low single arm BG from rad. Mott ($\times 0.1$ of the BG from Fe)
 - high beam currents allowed: continuous measurement

Operation:

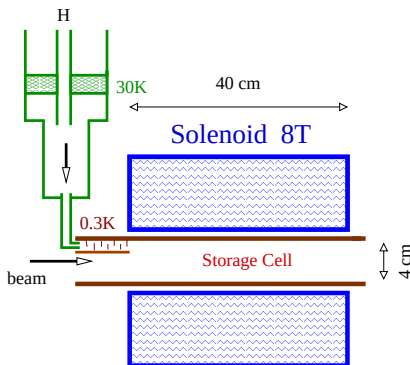
- density: $\sim 6 \cdot 10^{16}$ atoms/cm²
- Stat. error at 50 μ A: 1% in ~ 10 min

Møller Systematic Errors

Proposed: 100%-polarized atomic hydrogen target ($\sim 3 \cdot 10^{16}$ atoms/cm²).

Variable	Hall C	Hall A		
		Present	Upgrade	Proposed
Target polarization	0.25%	2.00%	0.50%	0.01%
Target angle	0.00%	0.50%	0.00%	0.00%
Analyzing power	0.24%	0.30%	0.30%	0.10%
Levchuk effect	0.30%	0.20%	0.20%	0.00%
Target temperature	0.05%	0.00%	0.02%	0.00%
Dead time	-	0.30%	0.30%	0.10%
Background	-	0.30%	0.30%	0.10%
Others	0.10%	0.30%	0.30%	0.30%?
Beam extapolation	?	larger	?	0.00%
Total	0.47%	2.10%	~0.80%	~0.35%

Storage Cell



First: 1980 (I.Silvera,J.Walraven)

$\vec{p}$ jet (Michigan)

Never put in high power beam

- 4 states, $\mathcal{P}_e$: $|a\rangle, |b\rangle = -1$ $|c\rangle, |d\rangle = +1$
- $-\vec{\nabla}(\vec{\mu}_H \vec{B})$ force in the field gradient
 - pulls $|a\rangle, |b\rangle$ into the strong field
 - repels $|c\rangle, |d\rangle$ out of the field
 - $\mathcal{P}_e = 1 - \delta$, $\delta \sim 10^{-5}$
- $H+H \rightarrow H_2$ recombination (+4.5 eV)
high rate at low T
 - parallel electron spins: suppressed
 - gas: 2-body kinematic suppression
 - gas: 3-body density suppression
 - surface: strong unless coated
 ~ 50 nm of superfluid ^4He
- Density $3 \cdot 10^{15} - 3 \cdot 10^{17} \text{ cm}^{-3}$.
- Gas lifetime > 1 h.

Contaminations and Depolarization of the Target Gas

Ideally, the trapped gas polarization is nearly 100% ($\sim 10^{-5}$ contamination).
 Good understanding of the gas properties (without beam).

Contamination and Depolarization

No Beam

Gas Properties

- Atom velocity ≈ 80 m/s
- Atomic collisions $\approx 1.4 \cdot 10^5$ s $^{-1}$
- Mean free path $\lambda \approx 0.6$ mm
- Wall collision time $t_R \approx 2$ ms
- Escape (10cm drift) $t_{es} \approx 1.4$ s

CEBAF Beam

- Bunch length $\sigma = 0.5$ ps
- Repetition rate 497 MHz
- Beam spot diameter ~ 0.2 mm

- Hydrogen molecules $\sim 10^{-5}$
- Upper states $|c\rangle$ and $|d\rangle < 10^{-5}$
- Excited states $< 10^{-5}$
- Helium and residual gas $< 0.1\%$
 - measurable with the beam

100 μ A Beam

- Depolarization by beam RF $< 2 \cdot 10^{-4}$
- Ion, electron contamination $< 10^{-5}$
- Excited states $< 10^{-5}$
- Ionization heating $< 10^{-10}$

Expected depolarization $< 2 \cdot 10^{-4}$

Potential Improvement of Systematic Accuracy

Fe at 3T: potential improvement (quite optimistic):

- Better understanding of magnetization in thin foils (find experts)
- More extensive MC and beam studies
- Measurements: 0.1% stat $\Rightarrow$ 3-5h beamtime $\Rightarrow$ beam stability?

Variable	Hall C	Hall A		
		Fe at 3T	H ₁ gas	
Target polarization	0.25%	0.50%	0.25%	0.01%
Target angle	0.00%	0.00%	0.00%	0.00%
Analyzing power	0.24%	0.30%	0.20%	0.15%
Levchuk effect	0.30%	0.20%	0.20%	0.00%
Target temperature	0.05%	0.02%	0.02%	0.00%
Dead time	-	0.30%	0.15%	0.10%
Background	-	0.30%	0.15%	0.10%
Others	0.10%	0.30%	0.15%	0.15%
Beam extapolation	?	0.15%	0.15%	0.00%
Total	0.47%	0.82%	0.48%	0.25%

Conclusion

New PV experiments require a $\sim 0.4\%$ polarimetry.
Two options for the Møller polarimetry:

Iron foil in strong field

- Not continuous
- Invasive
- Certain: 0.8%
- Potential: 0.5% with R&D
 - manpower ~ 4 FTE*Y
 - low material cost ($\sim 50k$)

Atomic hydrogen

- Continuous
- Not invasive
- Novel instrument
- Potential: 0.25% !
- Possible steps:
 - R&D - cell with electrodes
1 FTE*Y, 50k
- Interest expressed:
 - UVA (Don Crabb)
 - Mainz (Frank Maas)